

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)		Force per unit extension or $\frac{\text{force}}{\text{extension}}$ or $\frac{F}{x}$ (or e , Δl - not l unless stated as extension) Don't accept $F = kx$ without terms explained	1			1		
	(b)		Extension = 0.07 [m] (1) Spring force = $20 \times 0.07 = 1.4$ [N] (ecf for e) (1) Principle of Moments correct: $1.4 \times 0.9 = (1.2 \times 0.5) + (3x)$ (1) $x = 0.22$ [m] ecf for use of 0.45 (1) If $x = 0.24$ [m] award 3 marks only		4		4	3	
	(c)		Extension doubled for given force (can be implied) (1), so spring constant halved, [Laura correct] (1) Alternative: Extension = 14 [cm], force = 1.4 [N] (1) $k = \frac{1.4}{0.14} = 10$ [N m ⁻¹ so Laura is correct] (1)			2	2		
			Question 1 total	1	4	2	7	3	0